Duc Toan Nguyen

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Education

Aug 2025 – Present **Ph.D. in Electrical and Computer Engineering (ECE)**
Rice University, Houston, TX

May 2025 **Bachelor of Science in Mathematics**
Bachelor of Science in Computer Science
Texas Christian University (TCU), Fort Worth, TX
College of Science and Engineering, John V. Roach Honors College

Industry Experience

KLA Corporation, Ann Arbor May 2026 – Aug 2026

- **Role:** Algorithm Engineer Intern
- **Project title:** Topological and Geometric Representation for Robust Alignment and Reticle Inspection

– Proposed a **novel** anomaly-score heat map to localize scarce misalignment regions in tiny artifacts on **semiconductor** optical reticle images, supporting **semiconductor manufacturing** quality control

– Built an efficient **deep learning** model to classify misalignment accurately under strict speed and memory constraints for real-time **reticle inspection**

– Conducted experiments evaluating **state-of-the-art** image-processing methods, including **Persistent Homology** and **Morse-Smale representation**, for robust **semiconductor** defect detection

Research Experience

Rice University, Dept. of Electrical and Computer Engineering Feb 2026 – Present

- **Role:** Graduate Researcher (Decentralized Optimization on Riemannian Manifolds)
- **Advisor:** Dr. César A. Uribe

– Developed **the first intrinsic decentralized Riemannian optimization framework** achieving **exact convergence** in both consensus and objective error, matching the convergence rates of Euclidean decentralized optimization for smooth convex, nonconvex, deterministic, and stochastic settings.

– Developed **intrinsic** consensus analysis and geometric contraction techniques without relying on ambient Euclidean embeddings.

– Demonstrated the framework on Hadamard, sphere, Grassmann, and Bures–Wasserstein manifolds with applications to Fréchet mean, Fréchet median, and related manifold optimization problems.

Rice University, Dept. of Electrical and Computer Engineering Aug 2025 – Present

- **Role:** Graduate Researcher (Fréchet Regression on the Bures-Wasserstein Manifold)
- **Advisor:** Dr. César A. Uribe

– Leverage the intrinsic Riemannian geometry of the **Bures-Wasserstein metric** to design efficient optimization algorithms for **network regression** on graph Laplacians.

– Derive rigorous **bottleneck conditions** characterizing stationary points of the **Fréchet mean** optimization problem with Wasserstein metric.

– Apply network regression methods to real-world datasets, specifically modeling **diffusion tensor MRI**, to advance computational methods in **neuroscience**.

Independent Research Apr 2024 – May 2025

- **Role:** Researcher (Variance-Reduced Shuffling Stochastic Optimization)
- **Advisors:** Dr. Lam M. Nguyen (IBM Research), Dr. Trang H. Tran (Emory University)

– Propose a **novel variance-reduced optimization method** by applying shuffling paradigms to SARAH, creating a protocol that is practically efficient for **large-scale machine learning** workloads.

– Prove a **faster convergence rate** for the proposed method compared to existing state-of-the-art algorithms, directly addressing **AI computational efficiency**.

– Develop an inexact version for Expectation Maximization (EM) problems with rigorous proofs of complexity comparable to leading stochastic methods.

Texas Christian University, Dept. of Computer Science Jan 2023 – May 2025

- **Role:** Research Assistant (GO2AI Project)
- **Advisors:** Dr. Liran Ma, Dr. Ze-li Dou

– Implement Monte Carlo Tree Search (MCTS) and Convolutional Neural Networks (CNNs) into the policy architecture of distributed **AI agents** (Go).

– Optimize **reinforcement learning** algorithms by enhancing the training pipeline through distributed learning techniques.

– Implement **Grad-CAM (Explainable AI/XAI)** to visualize and interpret black-box learning processes, evolving the transparency of AI decision-making.

Texas Christian University, Dept. of Mathematics Sep 2022 – May 2025

- **Role:** Honors Research Scholar (Geodesic Nets - Construction and Existence)
- **Advisor:** Dr. Ken Richardson

– Develop a **first-of-its-kind heuristic algorithm** to construct approximate Steiner Trees for arbitrary point sets, **reducing total network length by 2%** compared to Minimum Spanning Trees.

– Apply the algorithm to optimize network architecture designs, with direct implications for efficient **communication infrastructures** and **genetic network analysis**.

– Prove theoretical conditions for the existence of Fermat points in triangles on general 2-D surfaces.

Rice University, Dept. of Statistics (REU) May 2023 – Jul 2023

- **Role:** Undergraduate Researcher (STAT-DATASCI REU 2023)
- **Advisor:** Dr. Eric C. Chi

– Investigate the sensitivity of optimal tuning parameters to noise levels in Nonnegative Matrix Factorization (NMF), a critical technique for **dimensional reduction**.

– Spearhead the development of a **novel algorithm, "Square-Root Min-Vol NMF,"** and establish its **rigorous convergence guarantees**.

– Validate the algorithm on large-scale datasets of **hyperspectral images**, demonstrating superior error reduction compared to recent methods, with applications in **biomedical imaging**.

Academic Publications

- **Nguyen, D. T.**, & Uribe, C. A. "Fréchet Regression on the Bures-Wasserstein Manifold." Neural Information Processing Systems (NeurIPS) 2026.
- **Nguyen, D. T.**, & Uribe, C. A. "On the Finite-Time Analysis for Intrinsic Decentralized Riemannian Optimization." NeurIPS Optimization for Machine Learning (OPT) Workshop 2026.
- **Nguyen, D. T.**, & Uribe, C. A. "Intrinsic Decentralized Stochastic Riemannian Optimization on Manifolds with Bounded Sectional Curvature". IEEE Control Systems Letters (2026). Invited for presentation at the *IEEE Conference on Decision and Control (CDC 2026)*.
- **Nguyen, D. T.**, Tran, T. H., & Nguyen, L. M. "Adjusted Shuffling SARAH: Advancing Complexity Analysis via Dynamic Gradient Weighting". Journal of Optimization Theory and Applications, 210(1), (2026): 12.
- **Nguyen, D. T.** "On the existence of a balanced vertex in geodesic nets with three boundary vertices." *Journal of Geometry*, 116.3 (2025): 36.
- **Nguyen, D. T.**, & Chi, E. C. "Towards tuning-free minimum-volume nonnegative matrix factorization." *Proceedings of the 2024 SIAM International Conference on Data Mining (SDM24)*.
- **Nguyen, D. T.** "Geodesic Nets - Construction and Existence." *(Outstanding Honors Thesis)*.

Honors & Awards

2026 **Student Travel Funding** – IEEE Control System Society (CSS) for IEEE CDC 2026
2026 **Pi Mu Epsilon Outstanding Speaker Award** – Joint Mathematics Meetings (JMM)
2025 **Senior Scholar** – Department of Mathematics, TCU
2025 **Best Undergraduate Poster** – TCU Student Research Symposium
2025 Finalist, **Best Honors Thesis Presentation** (Boller Competition) – TCU
2025 **Student Travel Funding** – Pi Mu Epsilon (PME) for JMM 2025
2025 Member, **Pi Mu Epsilon** (TCU Texas Alpha chapter)
2025 Member, **Upsilon Pi Epsilon** (TCU chapter)
2024 **Top 300** (of 3,988 students) – 85th William Lowell Putnam Mathematical Competition
2024 **SIAM Student Travel Award** – SIAM Int. Conference on Data Mining (SDM24)
2023 **Outstanding Session Presentation** – Comp. Math & Operations Research, GCURS
2022 **Top 500** (of 3,415 students) – 83rd William Lowell Putnam Mathematical Competition

Professional Activities

- **Reviewer**, *International Conference on Learning Representations (ICLR) 2026.*
- **Reviewer**, *The Conference on Neural Information Processing Systems (NeurIPS) 2026.*
- **Reviewer**, *Journal of Machine Learning Research (JMLR).*
- **Reviewer**, *Journal of Optimization Theory and Applications (JOTA).*
- **Reviewer**, *Statistica Sinica.*
- **Reviewer**, *IEEE Transactions on Signal and Information Processing over Networks.*
- **Reviewer**, *Computational Optimization and Applications.*
- **Reviewer**, *Discrete & Computational Geometry.*

Posters & Presentations

- **Nguyen, D. T.**, & Uribe, C. A. "Fréchet Regression on the Bures-Wasserstein Manifolds." *SIAM Conference on Optimization*, Jun 2026.
- **Nguyen, D. T.**, & Uribe, C. A. "Fréchet Regression on the Bures-Wasserstein Manifolds." *INFORMS Conference on Optimization*, Mar 2026.
- **Nguyen, D. T.** "Geodesic Nets - Construction and Existence." *PME Contributed Session on Research by Undergraduates*, JMM, Jan 2026.
- **Nguyen, D. T.**, & Chi, E. C. "Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization." *AMS Contributed Papers Session, Numerical Analysis I*, JMM, Jan 2025.
- **Nguyen, D. T.** "On the existence of a balanced vertex in geodesic nets with three boundary vertices." *AMS - PME Undergraduate Student Poster Session*, JMM, Jan 2025.
- **Nguyen, D. T.**, & Chi, E. C. "Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization." *SIAM Conference on Mathematics of Data Science (MDS24)*, Oct 2024. (Poster).
- **Nguyen, D. T.** "Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization." *Summer school on Bayesian learning and network analysis*, VIASM, Hanoi, Jul 2024. (Poster).
- **Nguyen, D. T.**, & Chi, E. C. "Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization." *SIAM International Conference on Data Mining (SDM24)*, Apr 2024. (Slides/Poster).
- **Nguyen, D. T.** "A Majorization-Minimization Variant For Minimum-Volume Nonnegative Matrix Factorization." *National Collegiate Research Conference (NCRC)*, Harvard University, Jan 2024.
- **Nguyen, D. T.** "Towards Tuning-Free Minimum-Volume Nonnegative Matrix Factorization." *Gulf Coast Undergraduate Research Symposium (GCURS)*, Rice University, Oct 2023.
- **Nguyen, D. T.** "Searching for networks of minimum length." *Research and Creative Activities Week*, TCU, Sep 2023. (Poster).
- Leath, H., Good, B., Fahimi, S., **Nguyen, D. T.**, Ma, L., & Dou, Z. "The Sybil in AI: The Many Personalities of a Go Playing Model." *Research and Creative Activities Week*, TCU, Sep 2023. (Poster).
- **Nguyen, D. T.** "Geodesic Nets construction using Genetic Algorithm." *Student Research Symposium (SRS)*, TCU, Apr 2023. (Poster).

Technical Skills

Languages	Python, Java, MATLAB, R, MySQL, C, C++, HTML, JS, PHP
AI/ML	PyTorch, Scikit-learn, NetworkX, NumPy, Pandas, Captum, TensorFlow
Systems	Linux, MacOS, Windows, Distributed Computing